OneBridge Grid: Product Requirements Document v1.1

Revision of v1.0 (3 October 2026) · 9 October 2026 · Draft for review · Prepared with Claude

1. About this revision

v1.1 records the decisions made after the v1.0 review and replaces the sections they affect. Every section of v1.0 that is not listed below is unchanged and still applies (Appendix A lists them). **[Assumption]** marks a proposal that is not yet validated. **[Pending approval]** marks a review recommendation the founders have not yet accepted.

#	Decision or change	Replaces in v1.0	Where in v1.1
D1	Multi-repo engineering model. No monorepo	§17.1, §17.2, §21	§5, §6
D2	No general Free plan. Free access for verified students only (college ID plus college email)	§6.1, §11.1, §18.1, F1, P1-BILL-01	§4, §10
D3	Offboarding: a departing person keeps only their own public card information	F3, Q4	§4.3
D4	Dashboard is one repo with feature modules and CODEOWNERS	§17.1	§6.8
D5	PWA fully specified	P1-CARD-08	§7
D6	UI/UX design foundations derived from the logo	new	§8, §9
D7	Review findings folded into scope	§5, §11	§3
D8	Open questions answered or re-assigned	§24	§13

2. Executive summary (revised)

OneBridge Grid remains a Digital Business Identity and Relationship Infrastructure: the company owns the identity, the cards and the relationships, and employees are custodians. Four things change in v1.1.

- Engineering moves to a multi-repo model. Each repo has one owning squad, and squads talk to each other only through versioned contracts. There is no monorepo.
- The Free plan is replaced by free access for verified students only. Every other customer pays.
- Offboarding is strict. A departing employee keeps only their own public card information.
- The PWA is specified in full, including what it cannot do on iOS, and the UI/UX direction is derived from the logo: an orange arch bridge with a person at its end.

Phase order, the North Star metric (MABR) and the India-first positioning are unchanged.

3. Review findings and scope adjustments

The v1.0 review raised the points below. Every recommendation is [Pending approval] until the founders decide.

#	Finding	Recommendation
1	The pitch (relationship infrastructure) is bigger than what Phase 1 ships (a very good digital card), so buyers will compare on price	Make the handover the hero moment of Phase 1. When a seat is offboarded the admin sees that leads, saved contacts and card history stay with the company (P1-HAND-01, Section 4.4)
2	Phase 1 Must scope is heavy for 8–10 people in 5 months	Defer the custom template builder (keep the 8 premium templates), wallet passes and auto-translate until after GA. Ship the student plan as a Should
3	Relationships are the core, but they live in MongoDB with $graphLookup	Keep the graph behind the Network service API from its first release, so the store can change at Phase 3 without touching consumers
4	The company owns the contacts, so the employee has no personal incentive to share cards	Give employees their own value: instant new-lead push, a 24 h follow-up reminder (lite version pulled forward from P2-CRM-03), a personal stats view and a premium card (P1-HAND-02)
5	MABR cannot be measured well before Phase 2	Judge Phase 1 on activation, view-to-save rate and seat utilisation. Report MABR from Phase 2

#	Finding	Recommendation
6	Open questions Q1 (GRID AI) and Q2 (truncated deck) weaken positioning and sourcing	See Section 13
7	A Free tier with OneBridge branding on every card clashes with 'premium by default'	Superseded by D2. The student card carries a refined keystone badge instead of a banner

4. Revised Phase 1 requirements

4.1 Student plan (new, replaces the Free plan)

Students are individuals, so they do not fit the rule that the company owns the identity. A student gets a **personal profile owned by the student**, with one card. It is separate from company seats, and it is the same person-owned identity that Phase 4 (OneBridge Career) needs, so the work carries forward.

ID	Requirement	Pri.	Acceptance criteria
P1-STU-01	Student account: a person-owned personal profile with one card, not tied to a company	S	No organisation features. ID format OB-S<6 char base32> [Assumption]
P1-STU-02	Eligibility: current student of a recognised institution. Expected graduation year captured at signup	S	Self-declared, then confirmed by STU-03 and STU-04
P1-STU-03	College email verification: OTP sent to an institutional email address	S	Domain checked against an institution allowlist managed in the ops console. Free-mail domains rejected
P1-STU-04	College ID verification: upload of the student ID card (front, and back if needed)	S	Ops review queue, 48 h SLA [Assumption]. Approve or reject with reason. Decision logged
P1-STU-05	Both checks are required by default. For institutions with no official email domain, ID plus a second proof (bonafide letter or fee receipt) is accepted	S	Fallback cases are flagged for senior review. See OPEN-B

ID	Requirement	Pri.	Acceptance criteria
P1-STU-06	Fraud and privacy controls	S	One active student plan per person (phone number plus ID hash). Rate limits. ID images encrypted and deleted 30 days after the decision [Assumption]. DPDP notice and consent recorded
P1-STU-07	Annual re-verification and lapse handling	S	Reminder 30 days before expiry. On lapse the card stays live read-only for 30 days, then shows a basic profile. One-click upgrade to Starter. Data is never deleted silently
P1-STU-08	Plan limits	S	1 card, 2 templates, 5 catalogue or portfolio items, 7-day analytics, share-back notifications by email only, OneBridge keystone badge
P1-STU-09	Ownership and portability	S	The student owns the profile. Export and delete (DPDP). No company data. Becomes the base of the Phase 4 career profile. If the student later joins a company, a separate seat is created and the personal card stays personal

The v1.0 non-goal 'consumer cards for non-business users in Phases 1–3' is amended: verified students are the only exception.

4.2 F1 Company onboarding (revised)

- Sign up with mobile OTP or Google/Microsoft. Enter company name, industry, size, city and GSTIN (optional at signup, required for the verification badge). Students choose 'I am a student' and follow Section 4.1.
- Choose a plan: Starter or Business (Enterprise = contact sales). Pay through Razorpay (UPI, cards, netbanking). There is no Free plan.
- The 14-day Business trial in v1.0 is on hold pending OPEN-A. Until the founders decide, treat it as removed.
- Brand setup: logo, colours, cover image, with a live card preview that updates as the user types.

- Add the first employee (self), publish the card, then prompt to share via WhatsApp and to invite colleagues (CSV/Excel import, invite link, or HRMS sync on Enterprise). Target: first card in under 10 minutes.

4.3 F3 Employee offboarding (revised, closes Q4)

- The admin marks the employee as leaving and chooses: (a) deactivate the card with a 'no longer with the company, contact X' page, (b) redirect to the successor's card, or (c) redirect to the department or company page.
- The person's login access to company data is revoked immediately (within 1 minute).
- The seat, its leads, relationships, saved-contact history and analytics stay with the company and transfer to the successor or manager.
- **The person keeps only their own public card information**: name, designation and the contact details shown on their public card, as a personal export. No leads, no contacts they collected, no notes, no analytics.
- The NFC destination is re-pointed through the short link, with no re-encoding. Hardware reissue is optional.
- When an employee claims a seat, a plain-language notice explains this rule and the acceptance is recorded (DPDP). This answers the 'company tracks my contacts' risk.
- The v1.0 admin override for personal copies is removed.

4.4 Handover summary and employee value (new) [Pending approval]

ID	Requirement	Pri.	Acceptance criteria
P1-HAND-01	At the end of F3 the admin sees a handover summary	S	Shows leads, saved contacts, share-back submissions and card views that stay with the company, the successor, and the message visitors will see. Also emailed to the admin and logged in the audit log
P1-HAND-02	Employee value pack	S	New-lead push in under 10 s. Reminder when a lead is untouched for 24 h (push, WhatsApp or email, opt-out supported). Personal stats view. Replaced by full CRM reminders at P2-CRM-03

4.5 Phase 1 MVP cut (revised)

All Must requirements in v1.0 Section 11, except the items deferred in Section 3 (custom template builder in P1-CARD-05, wallet passes in P1-CARD-07, auto-translate in P1-CARD-09). The student plan (P1-STU-*) and the handover items (P1-HAND-*) are Should and are scheduled in the last month before GA. The PWA (Section 7) stays a Must.

5. Working model: teams, repos and ways of working

5.1 Principles

- One repo is one deployable thing with one owning squad.
- Squads integrate only through versioned contracts (APIs and events), never through shared database access or shared source folders.
- Each service owns its own database.
- Shared code is published as versioned packages, not copied.
- Unfinished work merges behind feature flags, so no squad blocks another.
- Every cross-squad decision is written down as an RFC or ADR.

5.2 Squads and ownership

With 8–10 people in Phase 1, a squad may be 1–3 people. Ownership still follows this table.

Squad	Owns (repos)	Owns (in the dashboard repo)
Platform and Design System	ob-platform-infra, ob-ui-kit, ob-api-contracts, ob-e2e-tests	The app shell
Identity and Org	ob-svc-identity, ob-svc-organization	employees, org-structure
Cards and NFC	ob-svc-cards, ob-edge-shortlink, ob-web-public	cards, nfc-orders, my-card
Billing and Ops	ob-svc-payments, ob-svc-notifications, ob-admin-console	billing
Insights	ob-svc-analytics	analytics

5.3 Rules inside every repo

- Trunk-based development. Branches live 2 days or less and merge by squash.

- Protected main branch. A CODEOWNERS file names the owning squad. Every PR needs at least one owner review and green CI.
- Conventional Commits, with automatic semantic versioning and changelogs (semantic-release or Changesets).
- A preview environment for every PR, then dev, staging and production promotion.
- Feature flags for anything not ready for users.
- Renovate or Dependabot opens upgrade PRs for shared packages, so consumers upgrade on their own schedule.

5.4 Changing something that spans repos

1. Propose the change in ob-api-contracts. Consumer squads review the PR.
2. Expand: add the new field or endpoint without removing the old one.
3. Publish the new generated API client.
4. Each consumer migrates in its own repo and its own time.
5. Contract: remove the old field only after all consumers have moved and the deprecation window (two releases) has passed.
6. Breaking changes go in a new API version (/v2).

Changes that touch more than one squad's repo need a short RFC in ob-architecture-docs. A 30-minute weekly architecture sync reviews open RFCs.

5.5 Definition of done

- Contract updated and contract tests green (consumer-driven, Pact).
- Unit and integration tests green. Affected end-to-end tests green.
- Telemetry added (traces, structured logs, product events from Section 19 of v1.0).
- Accessibility checked for UI changes. Docs and runbook updated.
- Behind a flag if user-visible and not yet released.

5.6 Known costs of multi-repo, and the mitigation

Cost	Mitigation
Version drift between services and shared packages	Automated upgrade PRs, a weekly dependency report, a maximum of two minor versions behind
A change that spans three repos needs three PRs in order	Contract-first process (5.4), expand-then-contract migrations

Cost	Mitigation
Duplicated tooling and CI setup	ob-platform-infra provides shared CI templates, linting and base images
Hard to see the whole system	ob-architecture-docs holds a service map, event catalogue and runbooks
Slower bootstrap in the first weeks	Platform squad ships a repo template and CI pipeline before feature work starts [Assumption: about 2–3 weeks of platform work]

6. Technical architecture (replaces v1.0 Section 17)

6.1 Principles

- Edge first. Public card pages and short-link redirects are served from the edge cache, so QR and NFC taps work even if the origin is down.
- Squad-aligned services. Few services, each owned by one squad. Modular, not a swarm of micro-services.
- Database per service. Services never read each other's data directly. They use APIs or events.
- India data residency (AWS Mumbai primary, Hyderabad for DR), as in v1.0.
- Observable and secure by default.

6.2 Repository catalogue

Repo	Purpose	Stack	Phase
ob-web-public	Visitor card pages and public company profiles (SSR/ISR at the edge)	Next.js, TypeScript	1
ob-web-app	Dashboard for admins and employees, plus the My Card PWA	React/Next.js, TypeScript	1
ob-admin-console	Internal ops: KYB queue, student verification queue, NFC orders, moderation	React, TypeScript	1

Repo	Purpose	Stack	Phase
ob-ui-kit	Design tokens, components and icons, published as @onebridge/ui	React, Storybook	1
ob-svc-identity	Auth, OneBridge Business ID, KYB, student verification, RBAC	NestJS, MongoDB	1
ob-svc-organization	Company, branches, departments, seats, employees, brand kit, audit log	NestJS, MongoDB	1
ob-svc-cards	Cards, templates, QR, NFC chip registry, short links, share-back	NestJS, MongoDB	1
ob-svc-payments	Plans, subscriptions, Razorpay, GST invoices, NFC orders	NestJS, relational database	1
ob-svc-notifications	Email, SMS (DLT), WhatsApp, web push, in-app	NestJS, Redis	1
ob-svc-analytics	Event ingestion, aggregation, dashboards	NestJS, ClickHouse	1
ob-edge-shortlink	QR/NFC redirect worker	Cloudflare Workers	1
ob-api-contracts	OpenAPI specs, event schemas, generated clients	YAML, JSON Schema	1
ob-platform-infra	Terraform, CI templates, gateway config, base images	Terraform, GitHub Actions	1
ob-e2e-tests	Cross-service end-to-end tests	Playwright	1
ob-architecture-docs	ADRs, RFCs, service map, runbooks	Markdown	1
ob-svc-network, ob-svc-crm, ob-svc-search	Connections and graph, CRM, directory search	NestJS, OpenSearch	2
ob-mobile-app	Native iOS/Android apps with NFC	React Native (Expo)	2
ob-svc-ai, ob-svc-marketplace	AI gateway and RAG, marketplace and RFQ	NestJS	3

Repo	Purpose	Stack	Phase
ob-svc-career	Career profiles, jobs, mentorship	NestJS	4

6.3 Runtime view

- Clients (visitor browser, My Card PWA, dashboard, ops console, later native apps) reach Cloudflare: CDN, WAF and Workers.
- Short-link requests are resolved at the edge worker and redirected. Card pages are served from the cached ISR output of ob-web-public.
- Authenticated API traffic goes through the API gateway (token validation, rate limits, routing, request IDs) to the services.
- Services publish and consume events over SNS topics and SQS queues. Each service writes only to its own database.
- Analytics events are sent in batches directly to the analytics service, not over the business event bus, because of their volume.

6.4 Technology stack

Layer	Choice	Note
Public pages	Next.js (App Router, ISR), TypeScript, Tailwind with @onebridge/ui tokens	LCP and size budgets in v1.0 Section 16.1
Dashboard and PWA	React with Next.js, TypeScript, Workbox (through Serwist)	See Section 7
Mobile (Phase 2)	React Native (Expo)	Shares the TypeScript API client
Backend	Node.js, NestJS, TypeScript, REST with OpenAPI 3.1	GraphQL public API in Phase 3
Primary data	MongoDB Atlas (Mumbai) for identity, organization, cards and notifications	tenantId on every document
Payments data	Relational database (PostgreSQL or MySQL with Prisma) [Assumption]	Strict transactions for invoices. Billing squad confirms the engine in an ADR
Cache, jobs	Redis, BullMQ	Sessions, rate limits, in-service jobs

Layer	Choice	Note
Messaging	AWS SNS and SQS	Move to Kafka only if volume requires it
Analytics	ClickHouse	Customer-facing analytics
Search (Phase 2)	OpenSearch or Meilisearch	Directory and people search
Media	S3 (ap-south-1), Cloudflare CDN and Images	Student ID images in a separate encrypted bucket
Edge and gateway	Cloudflare Workers, WAF. API gateway (Kong or AWS API Gateway) [Assumption]	Gateway choice by ADR
Auth	OTP through MSG91, Google and Microsoft OAuth, SAML/OIDC, short-lived JWT access tokens with rotating refresh tokens	MFA (TOTP) for admins
Delivery	Docker, AWS ECS (EKS later), Terraform, GitHub Actions	Shared CI templates
Observability	OpenTelemetry, Sentry, Grafana	Correlation ID on every request
Testing	Jest or Vitest, Playwright, Pact (contract tests), k6 (load)	
Code quality	TypeScript strict mode, ESLint with module-boundary rules, Prettier, commit linting	

6.5 Services, data ownership and events

Service	Owns	Publishes
Identity	Accounts, sessions, Business ID, KYB status, student verification, roles	user.registered, kyb.status_changed, student.verification_decided
Organization	Company, branches, departments, teams, seats, brand kit, audit log	seat.created, seat.assigned, seat.offboarded, org.branding_updated

Service	Owns	Publishes
Cards	Cards, templates, QR, short links, NFC chips, share-back submissions, student personal cards	card.published, nfc.chip_activated, nfc.chip_disabled, shareback.submitted
Payments	Plans, subscriptions, invoices, NFC orders	plan_limits.changed, invoice.issued, nfc.order_status_changed
Notifications	Templates, preferences, push subscriptions, delivery log	none
Analytics	Event store and aggregates	none

Event	Publisher	Main consumers
seat.offboarded	Organization	Identity (revoke access), Cards (re-point or deactivate card), Notifications, Analytics
seat.assigned	Organization	Cards, Identity, Notifications
org.branding_updated	Organization	Cards (re-render within 60 s)
kyb.status_changed	Identity	Organization, Cards (badge), Notifications
student.verification_decided	Identity	Cards (plan state and badge), Notifications
plan_limits.changed	Payments	Organization (seat limits), Cards (template limits)
nfc.order_status_changed	Payments	Notifications
nfc.chip_activated	Cards	Organization (audit), Notifications, Analytics
shareback.submitted	Cards	Notifications, Analytics (CRM in Phase 2)

6.6 Contract and integration rules

- Every service publishes an OpenAPI 3.1 spec to ob-api-contracts. Events use JSON Schema or AsyncAPI with a version field. Typed clients are generated and published to a private npm registry as @onebridge/api-client.
- Consumer-driven contract tests (Pact) run in the CI of every service. A breaking change fails the PR of the squad that made it.
- Write endpoints accept an idempotency key. Events are delivered at least once, so consumers must be idempotent.

- Services publish events through a transactional outbox, so a database write and its event never diverge.
- Errors use the problem+json format. Lists are paginated. Every request carries a correlation ID.
- A service may keep a local read model built from events (a denormalised copy). The owning service stays the source of truth.
- Compatibility: add, never rename. Deprecation window is two releases. Breaking changes go in a new API version.

6.7 Core data model (updated)

Entity	Owner	Key attributes
Organization	Organization	businessId, legalName, displayName, gstin, kybStatus, plan, brandKit, slug, domains
Branch, Department, Team	Organization	id, orgId, name, address, managerSeatId, status
Seat (employee identity)	Organization	seatId, orgId, title, branchId, deptId, teamId, managerSeatId, status, assignedPersonId
Person	Identity	userId, phone, email, name, authProviders, mfa
StudentVerification (new)	Identity	personId, institution, graduationYear, emailStatus, idStatus, reviewer, decision, expiresAt
Card	Cards	cardId, ownerType (seat or student), ownerId, templateId, sections, visibility, language
ShortLink	Cards	code, targetType, targetId, source, status
NfcChip	Cards	uid, shortLinkCode, material, orderId, seatId, status
ShareBack and Lead (Phase 1 lite)	Cards	submissionId, cardId, name, phone, email, note, consent, ownerSeatId, createdAt
Order	Payments	orderId, orgId, items, amount, gst, shipping, status timeline
Subscription, Invoice	Payments	plan, seats, cycle, status, razorpayIds, gstBreakup

Entity	Owner	Key attributes
AuditLog	Organization	actor, action, entity, before, after, ts, ip (immutable)
PushSubscription (new)	Notifications	personId, endpoint, keys, deviceLabel, createdAt, revokedAt
AnalyticsEvent	Analytics	type, seatId, orgId, source, ts, device, city, sessionId

Relationship (edge), Contact and Deal entities belong to the Network and CRM services from Phase 2.

6.8 Dashboard repo structure (decision D4)

ob-web-app is one repo, organised by feature module.

```
src/
  app/                shell: routing, layout, auth guard, module registry
  features/
    org-structure/    branches, departments, org chart
    employees/        import, invites, offboarding, handover summary
    cards/            templates, card editor
    nfc-orders/
    analytics/
    billing/
    my-card/          the PWA employee and student home
  shared/             ui-kit wrappers, hooks, api-client setup
```

```
# CODEOWNERS
/src/app/                  @onebridge/squad-platform
/src/features/employees/   @onebridge/squad-identity-org
/src/features/org-structure/ @onebridge/squad-identity-org
/src/features/cards/       @onebridge/squad-cards-nfc
/src/features/my-card/     @onebridge/squad-cards-nfc
/src/features/billing/     @onebridge/squad-billing-ops
/src/features/analytics/   @onebridge/squad-insights
```

- A feature module may not import from another feature. It imports only from shared/ or from a module's public index.ts. An ESLint boundaries rule fails the build otherwise.
- Each module registers its own routes and menu items through the module registry, so nobody edits a central route file.
- Each module has its own tests and Storybook stories. CI runs only the tests of the modules a PR touches.

- The shell (src/app) is thin, rarely changed and owned by one squad. It is the only hotspot.

6.9 Edge, gateway and short links

- Short-link resolution happens in an edge worker with a cached code-to-destination map. Target: 100 ms p95 and 99.99% availability, as in v1.0.
- Destinations can be re-pointed by an admin. Disabled and unactivated chips resolve to a neutral page that shows no personal data.
- ob-web-public revalidates a card page when the Cards service publishes card.published or the Organization service publishes org.branding_updated.
- The gateway validates tokens, applies per-plan rate limits and adds the request ID. Authorisation (tenant and role scope) is enforced again inside each service.

6.10 CI/CD and environments

- Pipeline per repo: lint and type-check, unit tests, contract tests, build, security scans (dependencies, static analysis, container image), deploy to a preview environment, integration tests, staging, production.
- Environments: preview (per PR, short-lived), dev, staging, production. Lower environments use synthetic data only.
- Services deploy independently. Production uses canary or blue-green releases with automatic rollback on error-rate or latency alarms.
- Database changes follow expand-then-contract so that old and new versions run side by side.
- Frontends ship behind flags. Platform tracks the four DORA metrics (deployment frequency, lead time, change failure rate, time to restore).

6.11 Observability and operations

- OpenTelemetry traces across services with the correlation ID. Structured JSON logs without personal data.
- Each service has dashboards for the golden signals and SLO alerts, and a runbook in ob-architecture-docs.
- Sentry for front-end and back-end errors. On-call rota per squad.
- Analytics pipeline failure must not break card pages. Events queue and replay (v1.0 Section 16.2).

6.12 Security and multi-tenancy (additions)

- Service-to-service calls use signed service tokens (or mTLS) and least-privilege IAM per service. Secrets live in a managed vault.
- tenantId is enforced in each service's repository layer. Every service CI runs cross-tenant isolation tests.
- Student ID images are stored in a separate encrypted bucket, readable only by the ops review role, and deleted 30 days after the decision [Assumption].
- The offline cache in the PWA is wiped on logout, revocation or seat reassignment (Section 7.7).
- All v1.0 security, privacy and compliance requirements (Section 16) still apply.

6.13 API surface additions

Endpoint	Purpose
POST /students/verify-email, POST /students/verify-email:confirm	Start and confirm college email OTP
POST /students/verify-id	Upload student ID for review
GET and PATCH /students/me	Student profile and plan state
POST /seats/{id}:offboard	Offboard (returns the handover summary)
GET /seats/{id}/handover-summary	Retrieve the handover summary
POST /push-subscriptions, DELETE /push-subscriptions/{id}	Register and revoke web push for a device
POST /leads:sync	Idempotent batch upload of leads captured offline

All v1.0 endpoints in Section 17.5 remain.

7. PWA specification (replaces P1-CARD-08)

v1.0 said 'PWA' in several places without defining it. This section defines what the PWA is, what it does offline, and what it cannot do on each platform.

7.1 Scope: which surface is a PWA

Surface	PWA?	Reason
Visitor card page	No	A fast, no-install web page. Asking visitors to install anything breaks the zero-friction principle
My Card (employees and students)	Yes. Installable, with offline QR and card	The daily-use surface: show my QR, share, my leads, my stats
Admin dashboard	Responsive web. Installable shell, online only	Needs live data. No offline mode
Ops console	No	Desktop web, internal

My Card lives in ob-web-app (feature module my-card) and is served from its own scope, so the service worker never touches the admin screens.

7.2 App identity and install

- Web app manifest: id, name 'OneBridge', short_name 'OneBridge', display standalone, portrait orientation, theme colour Bridge Orange, background Warm Paper (Section 8), start_url '/my-card?source=pwa', scope '/my-card/'.
- Icons: 192 and 512 px plus a maskable icon, using the arch glyph from the logo.
- Shortcuts: Show my QR, My leads, Share my card.
- Share target: receive a shared contact or link and offer to add it as a lead [Should].
- One manifest for all customers. The organisation's logo appears inside the app, not in the home-screen icon, because per-company manifests would complicate caching and updates [Assumption].
- Install prompt: Android and desktop Chromium show a custom prompt after the user's first successful card share, never in the first session before they have seen value. iOS has no install prompt, so show a short 'Share, then Add to Home Screen' guide with a screenshot. The user can dismiss it, and it returns after 14 days at most.

7.3 Service worker and caching

Resource	Strategy	Notes
App shell (JS, CSS, fonts, icons)	Precache, versioned with each release	Initial JS budget in Section 7.8

Resource	Strategy	Notes
My card data (GET /cards/me)	Stale-while-revalidate, stored in IndexedDB	Shows instantly, refreshes in the background
QR code	Generated on the device from the short link	Needs no network. Cached as SVG
Leads list	Network-first, falls back to the last 50 cached leads	Read-only offline
Images (photo, logo)	Cache-first with expiry (60 entries, 30 days)	
Writes (new lead, notes)	Queued in IndexedDB	See Section 7.4
Auth and billing endpoints	Network only	Never cached
Analytics events	Batched and queued	Never block the UI

- Updates: a new service worker waits. The app shows an 'Update available' banner and reloads only when the user taps it, or at the next cold start. A security-critical release can force an update.
- An offline fallback screen shows the QR and card if available, otherwise a clear 'You are offline' message.

7.4 Offline behaviour

Works offline	Needs a connection
Show my card and QR	Edit the card or profile
Share the link through the device share sheet (when the OS allows it)	Order NFC cards, billing
Browse the last 50 synced leads	See fresh analytics
Capture a lead manually (queued)	Receive new-lead pushes

- The queue stores each action with an idempotency key. It flushes when the app regains connectivity and every time the app opens, because iOS has no Background Sync. Retries use exponential backoff, and an action is dropped only after 7 days, with the user told.
- A visible indicator shows 'N changes waiting to sync' and the time of the last sync.

- Profile edits are not allowed offline in Phase 1, which avoids merge conflicts. Leads are append-only, so they cannot conflict.
- The QR keeps working whatever the sync state, because the short link is immutable.

7.5 Push notifications

- Standard Web Push with VAPID keys. The permission prompt appears in context (after the user first turns on lead alerts, or after their first lead), never on first load.
- Events from P1-NOT-02: new lead, approval requests, NFC order status, security alerts. Quiet hours and per-event preferences apply.
- Payloads are minimal: no phone or email, only a title and a deep link. Details load after the user opens the app.
- Target: new-lead notification delivered in under 10 s.
- iOS delivers web push only for an installed PWA on iOS 16.4 or later. Where push is unavailable or denied, the same event goes out by WhatsApp, SMS or email according to user preference.

7.6 Platform capability matrix

Capability	Android (Chrome)	iPhone (Safari, installed)	Desktop (Chrome, Edge)
Install	Custom prompt	Manual: Share, Add to Home Screen	Prompt or address-bar icon
Offline card and QR	Yes	Yes, once installed	Yes
Web push	Yes	Yes, iOS 16.4+ and installed only	Yes
Background Sync	Yes	No. Flush on open	Yes (Chromium)
Web NFC (read or write tags in the app)	Yes	No	No
Tap an NFC card to open a card page	Yes (system feature)	Yes (iPhone XS or later)	Not applicable
App badge	Where supported	Limited	Where supported
Storage reliability	Good	Reliable when installed. Can be cleared for sites that are not installed	Good

Browser behaviour changes between OS releases, so the device lab re-tests this matrix every release.

Consequence for the PRD: the PWA cannot read or write NFC tags on iPhones. In Phase 1 NFC encoding is done by ops, and employees activate a card by tapping it, which only opens a page and works on both platforms. In-app NFC (P2-MOB-01) needs the Phase 2 React Native app.

7.7 Security and privacy

- What is cached: the user's own card, the QR target, and the last 50 leads. Lead phone and email fields are encrypted in IndexedDB with a non-extractable WebCrypto key. The device lock is the main protection, and this is stated in the privacy notice.
- Wipe triggers: logout, session revocation, seat unassignment (offboarding), plan lapse, and 14 days without a successful sync [Assumption]. Online devices are wiped within about a minute of revocation through a push message and on the next request. Offline devices are wiped at the next connection or when the 14-day limit is reached. This residual window is accepted and documented.
- Tokens: short-lived access token kept in memory, refresh token in an httpOnly, Secure, SameSite cookie. Nothing sensitive in localStorage.
- Content Security Policy and subresource integrity on all scripts. The service worker handles same-origin requests only.
- Students follow the same rules. A student's data is deleted from the device on logout.

7.8 Performance and quality budgets

Metric	Target
Initial JavaScript for My Card	150 KB gzipped or less
Launch to QR visible, installed, offline	300 ms or less after the shell loads
LCP on Slow-4G, 2 GB RAM Android	2.0 s or less
Time to interactive, same conditions	3.5 s or less
Lighthouse	PWA installability checks all pass. Performance 90 or more. Accessibility 95 or more
Test devices	Android 9+ with 2 GB RAM, iPhone on iOS 15+ (push needs 16.4+), current desktop Chrome and Edge

7.9 PWA requirements

ID	Requirement	Pri.	Acceptance criteria
P1-PWA-01	Installable: manifest, service worker, HTTPS, icons	M	Passes Lighthouse installability. Installs on Android and iOS
P1-PWA-02	Offline card and QR	M	Works in airplane mode after first load and install
P1-PWA-03	Install UX: contextual prompt on Android, guide on iOS	M	Prompt after first share. Dismissal respected for 14 days
P1-PWA-04	Offline lead capture and sync queue	M	No data loss across reconnects. Idempotent. Pending count visible
P1-PWA-05	Web push for new leads and key events	M	Under 10 s delivery. Preferences and quiet hours honoured. Fallback channel on iOS and when denied
P1-PWA-06	Controlled update flow	M	Banner and user-initiated reload. Forced update path for security fixes
P1-PWA-07	Cache wipe on logout, revocation and offboarding	M	Verified in test: no lead data remains after wipe
P1-PWA-08	Shortcuts and share target	S	Shortcuts visible on Android. Share target opens add-lead
P1-PWA-09	App badge for unread leads	S	Where the platform supports it
P1-PWA-10	Device-lab test matrix run every release	M	Matrix in Section 7.6 passes or deviations are documented
P1-PWA-11	Native NFC read and write	Out of scope	Moved to P2-MOB-01 (React Native)

8. Design foundations (new)

8.1 Brand idea

The logo is an orange arch bridge with a person at its end, beside a charcoal pillar. The idea is a bridge that carries a person to the other side. The interface repeats that idea: arches for identity, arcs for connections, and a keystone for trust.

8.2 Colour tokens

Bridge Orange and Charcoal are sampled from the supplied logo image. Confirm them against the logo source file and lock all tokens in ob-ui-kit.

Token	Value	Use
Bridge Orange	#F68713	Primary action fills, highlights, the arch, key numbers
Charcoal	#393536	Text on orange, structure, headers, dark surfaces
Charcoal Deep	#1E1E1E	Dark-mode surfaces
Warm Paper	#FAF7F2	Page background (light mode)
Sand	#EFE8DC	Cards, dividers, subtle fills
Deep Orange	#A04E08	Orange used as text or icons on light backgrounds
Success	#27704A	Verified, synced, saved
Warning	#C99700	Untouched leads, pending sync
Error	#C2362B	Failures, destructive actions
Info	#3B6EA8	Neutral notices

Semantic colours are [Assumption] and need an accessibility check.

Contrast rules. Charcoal text on Bridge Orange is about 4.8:1 and passes. White text on Bridge Orange is about 2.5:1 and fails, so orange buttons always carry charcoal text. Orange text on a light background fails, so use Deep Orange (about 5.4:1 on Warm Paper). The earlier working value #B85A0A was too light for body text and is replaced.

8.3 Signature shapes

- **Arch top.** Cards, QR frames and avatars sit under an arch. Competitors use plain rounded rectangles.

- **Keystone badge.** The verification badge is a keystone at the crown of the arch. The student 'Made with OneBridge' badge uses a smaller version of the same keystone.
- **Arc connectors.** Connections, shares and handovers are drawn as arcs between two points. The relationship graph (Phase 2) uses arcs instead of straight lines.
- **Bridge-span progress.** Progress bars and loaders draw an arch span.

8.4 Typography

Manrope for headings and UI, with Noto Sans Devanagari, Telugu, Tamil, Kannada, Marathi, Bengali and Gujarati as script fallbacks [Assumption]. Fonts are self-hosted for speed and privacy. Numbers use Indian formatting (lakh, crore).

8.5 Four interface moods

User	Mood	Notes
Visitor	Warm, light, expressive	Thumb-zone actions. No login. Fast on low-end phones
Employee and student (My Card)	Quick, compact, dark-mode aware	Opens straight to the QR. One-hand use
Company admin	Calm, charcoal-led, data-dense	Orange only for the primary action and key numbers
Platform ops	Utilitarian	Queues, document viewers, audit trails. Desktop only

8.6 Motion

- When a visitor saves a contact or submits share-back, an arc draws between the two parties in about 400 ms, then settles.
- During offboarding, an arc carries the relationship counts from the leaving seat to the successor.
- Motion is decorative. All flows work without it, and 'reduce motion' replaces arcs with a simple fade.

8.7 Accessibility and content

- WCAG 2.1 AA for dashboards and public pages. Touch targets are 44 px or more.
- Plain, short copy. English and Hindi in Phase 1. Every screen has a visible next step.
- Do not rely on colour alone. Status chips always include an icon or word.

8.8 Component inventory (ob-ui-kit)

Arch card, keystone badge, action button (primary orange, secondary outline, ghost), QR frame, seat chip, stat tile, lead row, status chip, org-tree node, handover timeline, side drawer, empty state, toast, offline and update banners, install guide sheet.

8.9 States that must be designed

Loading skeleton, empty (no leads, no employees), error, offline, pending sync, unactivated NFC card, inactive or lost card, offboarded employee page, plan lapsed, student verification pending, student verification rejected, update available.

9. Reference screens (new)

Three reference images accompany this PRD as PNG files: ob-ui-01-visitor-card.png, ob-ui-02-my-card-pwa.png and ob-ui-03-admin-handover.png. They are references for design direction, not final pixel specifications. Names and numbers in them are illustrative.

9.1 Screen 1: Visitor card (mobile web)

- **Purpose:** turn a scan or tap into a saved contact and a lead, in one thumb reach.
- **Layout, top to bottom:** an orange arch header with the company logo, the keystone verification badge at its crown, the person's photo under the arch, then name, designation and company. Next come the primary actions: Save Contact (large, orange) and WhatsApp (outline), then Call, Email and Directions. Below sit the share-back panel ('Share your details', three fields and a consent checkbox), then About, Products, Portfolio, Gallery, Documents and social links. The footer carries the keystone 'Made with OneBridge' badge on student cards.
- **Interactions:** Save Contact downloads a vCard with photo and company. Share-back draws an arc between the visitor and the card owner and confirms.
- **States:** loading skeleton, language toggle (English and Hindi), unactivated NFC card, inactive card, offboarded page with 'contact X', slow connection.
- **Measures:** LCP 1.5 s or less on 4G. Card view to save rate 25% or more.

9.2 Screen 2: My Card home (installed PWA, mobile)

- **Purpose:** show my QR and share it in two taps, even offline.
- **Layout:** a header with a greeting, the organisation name and a sync chip ('Synced 2 min ago', or 'Offline'). The hero is a large QR inside an arch frame, with a source-tag selector and a full-screen button. Below it, a share row (WhatsApp, link, email signature, NFC

tap hint, hidden on iPhone) and three stat tiles (views, saves, leads, last 7 days). Then a leads list with a warning chip on leads untouched for 24 h and quick call and WhatsApp actions. A bottom bar holds Card, Leads, Stats and Profile.

- **States:** offline, 'N changes waiting to sync', update available, install prompt (Android) or install guide (iPhone), card not yet activated, notification permission request after the first lead.
- **Measures:** time to QR visible under 300 ms when installed. Seat utilisation 80% or more.

9.3 Screen 3: Admin dashboard and offboarding handover (desktop)

- **Purpose:** give admins a calm control room, and make the company-owned relationship story visible when someone leaves.
- **Layout:** a charcoal left rail (Dashboard, Organisation, Employees, Cards, NFC orders, Analytics, Billing), a top bar with the organisation switcher and search. The dashboard shows stat tiles (active cards against seats, views, saves, leads), a views-and-saves trend chart, a branch leaderboard and an activity feed.
- **Offboarding flow:** from Employees, choose 'Offboard'. A side drawer asks what visitors see (deactivate page, redirect to successor, redirect to company page) and who takes over. On confirmation the handover summary appears: the leaving seat and the successor joined by an arc, with the counts that stay with the company (leads, saved contacts, share-back submissions, card views), the visitor message, and a note that the person keeps only their public card information.
- **States:** bulk import preview with row errors, approval pending, over-limit seats read-only, plan lapsed banner, empty states.
- **Measures:** offboarding completed in under 2 minutes. Access revoked within 1 minute.

10. Plans and pricing (revised, replaces v1.0 Section 18.1)

The Free plan is removed. A Student plan replaces it. All prices exclude 18% GST and remain proposals [Assumption]. Hardware, add-on and AI pricing in v1.0 Section 18.2 is unchanged.

	Student	Starter	Business	Enterprise
Who	Verified students only	Small teams	Growing companies	Large, multi-branch companies
Price (per seat per month)	Free	₹199 (₹159 annual)	₹349 (₹279 annual)	Custom, from about ₹450, 100 seats minimum

	Student	Starter	Business	Enterprise
Seats	1 personal profile	1–25	10–1,000	100+
Owner of the identity	The student	The company	The company	The company
Digital card, QR, PWA	Yes, with keystone badge	Yes	Yes	Yes
Templates	2	All standard	All standard (builder after GA)	All plus bespoke design
Branches and departments	No	1 branch	Unlimited	Unlimited
Catalogue and portfolio items	5	50	Unlimited	Unlimited
Analytics retention	7 days	90 days	1 year	2+ years, exports, BI
Share-back lead capture	Email notification only	Yes	Yes	Yes
Admin approvals, bulk operations, audit log	No	Basic	Yes	Yes
KYB verification badge	Not applicable	Yes	Yes	Yes
SSO, SCIM, custom domain, custom roles	No	No	Add-on	Yes
CRM (Phase 2)	No	Add-on	Add-on	Included
Support	Community	Email	Priority chat	Dedicated manager, 99.95% SLA

The 14-day Business trial is on hold pending OPEN-A. The viral 'Get your OneBridge card' loop from v1.0 now runs from student cards and the student plan's keystone badge, so growth targets that assumed a Free plan must be re-based in the GTM plan.

11. Non-functional additions

Area	Requirement
PWA	Budgets and device support in Section 7.8
CI speed	Pull-request pipeline 15 min or less at p90 [Assumption]
Contract coverage	100% of public endpoints and events have contract tests
Code quality	New code at least 80% unit-test coverage [Assumption]. TypeScript strict mode. Module-boundary lint rule must pass
Dependencies	No repo more than two minor versions behind shared packages (ob-ui-kit, API client)
Delivery	Track DORA metrics per squad. Services deploy independently with automatic rollback
Student verification	Decision within 48 h of submission, 95% of cases [Assumption]
Offboarding	Access revoked within 1 minute. Handover summary within 5 seconds
Student ID data	Encrypted at rest, ops-review access only, deleted 30 days after decision [Assumption]
Analytics events added	student_verification_submitted, student_verification_decided, pwa_installed, push_permission_granted, offline_lead_queued, offline_lead_synced, handover_summary_viewed

All v1.0 non-functional requirements (Section 16) still apply.

12. Roadmap and risks (updates)

12.1 Phase 1 adjustments

The v1.0 windows stay as indicative [Assumption]: months 0–5 for Phase 1. Multi-repo adds platform work at the start.

Month	Milestone
M1	Design system and tokens, ob-api-contracts, repo template and shared CI, first ADRs. Identity and Organization skeletons
M2	Cards service, short-link edge worker, public card pages, dashboard shell with feature modules
M3	Internal alpha: onboarding, employees, cards, QR. NFC ordering started
M4	Closed beta (20–50 organisations): My Card PWA, analytics, billing and GST invoices, ops console
M5	GA. Handover summary and student plan (Should) land before GA if capacity allows

12.2 New and changed risks

Risk	Likelihood / Impact	Mitigation
Version drift between repos and shared packages	High / Medium	Automated upgrade PRs, weekly dependency report, maximum two minor versions behind
Slow cross-repo changes	Medium / Medium	Contract-first process, expand-then-contract, RFCs for cross-squad changes
Platform bootstrap delays feature work	Medium / Medium	Repo template and CI delivered in M1. Platform squad owns it
Fake or reused student IDs	Medium / Medium	Email plus ID checks, hash-based duplicate detection, ops review, yearly re-verification
Student verification queue overloads ops	Medium / Medium	48 h SLA monitoring, allowlist of institution domains to auto-approve email, reviewer capacity plan
Loss of the free-tier growth loop	High / Medium	Student loop and keystone badge, referral credits (P3-REF-02 pulled earlier if needed), event-led growth

Risk	Likelihood / Impact	Mitigation
iPhone limits (no Web NFC, no Background Sync, push only when installed)	High / Medium	Design around them (Section 7.6), flush queue on open, fallback channels, native app in Phase 2
Lead data left on a lost or offboarded device	Medium / High	Encrypted offline store, remote wipe on revocation, 14-day offline limit, device-lock guidance
Employee resistance to company-owned contacts	Medium / Medium	Plain-language notice at seat claim, visible privacy settings, employee value pack

All other v1.0 risks (Section 22) still apply.

13. Open questions (status)

#	Question	Status	Answer or next step	Owner
Q1	What is 'GRID AI' in the competitive table?	Proposed answer	Treat as a placeholder. Replace with named competitors (for example Blinq, Popl, HiHello) after verifying their current features and pricing	Founders, Product
Q2	The Ecosystem .pptx is truncated	Open	Re-export the deck and re-share it	Founders
Q3	Free tier or trial only?	Closed	Free access for verified students only, by college ID and college email (D2)	Done
Q4	What may a departing employee keep?	Closed	Only their own public card information (D3)	Done
Q5	Domains and legal entity name	Open	Check grid.onebridge.in and obg.in availability. Confirm the invoicing entity	Founders (month 1)

#	Question	Status	Answer or next step	Owner
Q6	Directory opt-in or opt-out	Deferred	Decide at Phase 2 kickoff	Product, Legal
Q7	Build or partner for CRM sync and HRMS	Proposed answer	Phase 2 starts with webhooks, CSV and Zapier or Make. Build one native HRMS connector, chosen by customer demand	Engineering
Q8	Marketplace revenue model	Proposed answer	Start with a fee per qualified RFQ lead. Add a success fee later	Founders
Q9	Phase 4 gating and brand	Deferred	Decide at the Phase 3 mid-point	Founders
Q10	Team size, budget, target GA date	Open	Needed to confirm Section 12	Management
OPEN-A	Does the 14-day Business trial stay for non-students?	Open	Currently treated as removed	Founders, GTM
OPEN-B	Student verification: are email and ID both mandatory, and is the no-email-domain fallback approved?	Open	Default is both required, with a documented fallback	Product
OPEN-C	Approve the Section 3 recommendations (defer builder, wallet passes, auto-translate; handover summary; employee value pack)	Open	Pending approval	Founders
OPEN-D	Single OneBridge manifest or per-company manifests for the PWA	Open	Default is a single manifest	Engineering, Design

Decisions to record as ADRs in the first month: repo and branching model, API gateway, payments database engine, messaging (SNS and SQS), and the PWA caching and wipe policy.

Appendix A. Carry-over index from v1.0

v1.0 section	Status in v1.1
1 Executive summary	Revised (Section 2)
2 Document control	Version and date updated. Sources unchanged
3 Problem statement	Unchanged
4 Vision, mission, positioning	Unchanged
5 Goals, non-goals, metrics	Unchanged, except the non-goal on consumer cards (verified students are an exception)
6 Market and personas	Free entry plan replaced by the Student plan. New persona: Student (a verified college student who wants a professional card)
7 Competitive positioning	Unchanged, pending Q1
8 Ecosystem and platform architecture	Unchanged. Student profile ID format added in P1-STU-01
9 Customer journeys and flows	F1 and F3 revised (Sections 4.2 and 4.3). F2 and F4 unchanged
10 Functional requirements overview	Unchanged
11 Phase 1 requirements	Revised by Sections 3, 4 and 7. P1-BILL-01 trial on hold (OPEN-A)
12–14 Phases 2–4	Unchanged
15 Roles and permissions	Unchanged. Add a personal Student role with own-profile scope only
16 Non-functional requirements	Unchanged, plus Section 11
17 Technical architecture	Replaced by Section 6
18 Monetisation and pricing	18.1 replaced by Section 10. 18.2 and 18.3 unchanged
19 Analytics plan	Unchanged, plus the events in Section 11
20 Go-to-market	Free-plan viral loop replaced by the student loop (Section 10)
21 Roadmap	Replaced by Section 12.1

v1.0 section	Status in v1.1
22 Risks	Extended by Section 12.2
23 Assumptions and dependencies	Unchanged
24 Open questions	Replaced by Section 13
25 Glossary	Extended by Appendix B

Appendix B. Glossary additions

Term	Definition
Multi-repo (polyrepo)	One repository per deployable unit, each owned by one squad
Squad	A small team that owns specific repos and modules end to end
Contract-first	Changing the API or event contract in ob-api-contracts before changing code
ADR	Architecture Decision Record: a short written record of a technical decision
RFC	Request for Comments: a proposal reviewed by affected squads before a cross-squad change
Expand then contract	Add the new field or endpoint first, migrate consumers, then remove the old one
CODEOWNERS	A repo file that assigns review ownership of folders to squads
PWA	Progressive Web App: a web app that can be installed and works offline
My Card	The installable PWA surface for employees and students
Student plan	Free plan for verified students, person-owned, with a keystone badge
Keystone badge	The arch-top verification mark used for verified companies and the student badge

Term	Definition
Handover summary	The screen shown to the admin after offboarding that lists what stays with the company